```
$ psql -c "SELECT ... FROM students JOIN courses ON ..."
```

SQL JOIN

// [illegible]

INNER	LEFT	RIGHT	FULL	CROSS
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

$ cat query-plan.md

□□□□

01 / □□ JOIN

□□□□□□□□□□□□□□□

02 / □□

□□□□□□□

03 / □□□□

ON vs WHERE □□□

04 / □□

□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□ = LEFT□□□ = INNER□

PART 01

□□ JOIN

□□□ ID: 1,2,3,4 × □□□ ID: 2,3,5

INNER JOIN□□□□

```sql
SELECT
    s.student_id,
    s.name AS student_name,
    c.course_name
FROM students AS s
INNER JOIN courses AS c
    ON s.student_id = c.student_id;
```

□□

□□□□□□□□□□□□□□——□□□□□□

RESULT SET

□□□ ID□1, 2, 3, 4

□□□ ID□2, 3, 5

→ □□□□ ID□2, 3

✗ □□□□□□□1, 4□□□□

✗ □□□□□□□5□□□□□

LEFT JOIN▯□□□□□□

```sql
SELECT
    s.name,
    c.course_name,
    CASE
        WHEN c.course_name IS NULL
        THEN '□□□' ELSE '□□□'
    END AS status
FROM students AS s
LEFT JOIN courses AS c
    ON s.student_id = c.student_id;
```

RESULT SET

→ □□□ID 1, 2, 3, 4 □□□□

→ 1 □ 4 □□□□□□□ NULL

□□

▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯——▯▯▯▯▯▯▯▯▯▯▯

LEFT JOIN □□□□□

□□ 1□□□□□□□□□

```sql
SELECT c.customer_name
FROM customers c
LEFT JOIN orders o
    ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
-- □□□□□□□
```

□□ 2□□□□□□□

```sql
SELECT s.name,
       COUNT(c.course_id) AS course_count
FROM students s
LEFT JOIN courses c
    ON s.student_id = c.student_id
GROUP BY s.student_id, s.name;
-- □□□□□□ count = 0
```

LEFT JOIN + IS NULL □□□□□□□□□□□□□□□——INNER JOIN □□□□

RIGHT□FULL OUTER□CROSS

RIGHT JOIN

□□□□□□□□□□□□

`A LEFT JOIN B` ≡ `B RIGHT JOIN`

`A` □□□□ LEFT □□□

FULL OUTER JOIN

□□□□□□□□□□ NULL□

MySQL □□□□□ LEFT ∪ RIGHT

□UNION□□□

CROSS JOIN

□□□□□3 □ × 4 □ = 12 □□

□□□□□□□□□□ × □□□□□□□

```sql
-- MySQL □□ FULL OUTER JOIN
SELECT * FROM students s LEFT JOIN courses c ON s.student_id = c.student_id
UNION
SELECT * FROM students s RIGHT JOIN courses c ON s.student_id = c.student_id;
```

PART 02

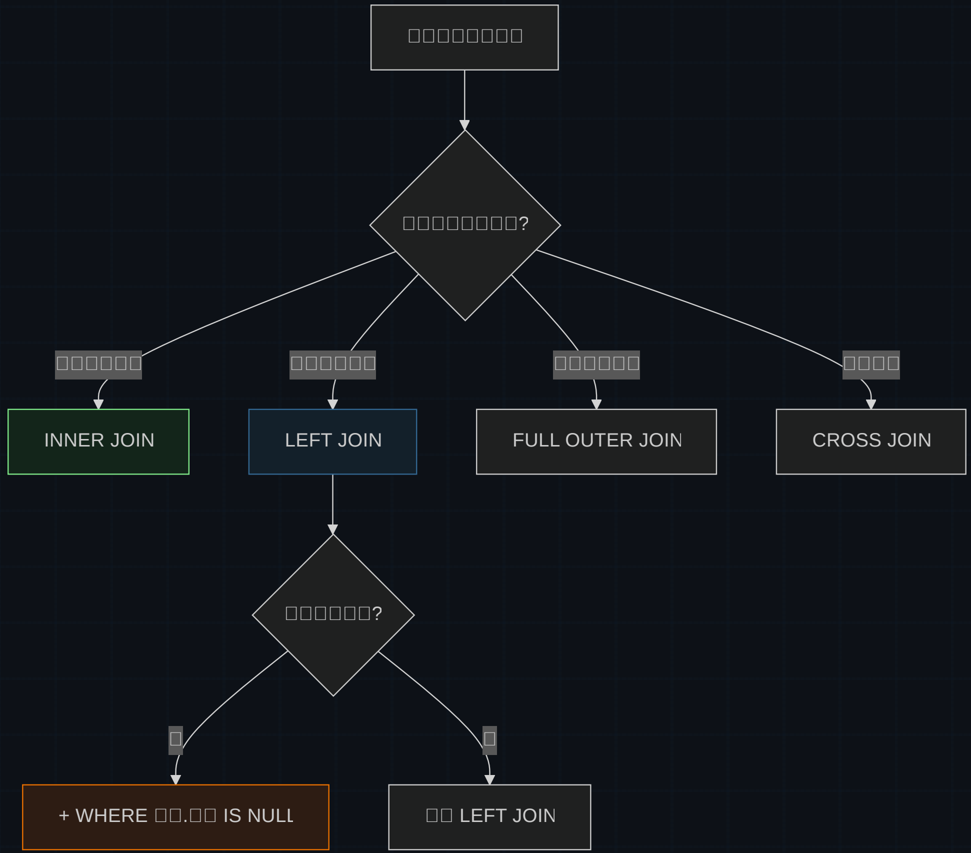
[illegible]

[illegible] + [illegible]

□□ JOIN □□

JOIN □□	□□	□□	□□□□	□□□□
INNER JOIN	□□□□	□□□□	□□□	⭐⭐⭐⭐⭐
LEFT JOIN	□□	□□□□	□□□ NULL	⭐⭐⭐⭐⭐
RIGHT JOIN	□□□□	□□	□□□ NULL	⭐⭐
FULL OUTER	□□	□□	□□□□ NULL	⭐⭐
CROSS JOIN	□□	□□	□□□□□□	⭐

```
$ 90% □□□□□□□ INNER □ LEFT —— □□□□□□□□
```


?
INNER JOIN
LEFT JOIN
FULL OUTER JOIN
CROSS JOIN
?
+ WHERE □□.□□ IS NULL
□□ LEFT JOIN

PART 03

□□□□□ON vs WHERE

90% □ LEFT JOIN bug □□□

CHECKPOINT_01

LEFT JOIN courses c ... **WHERE** c.course_name = 'Math' □□□□□

- A □□□□□□□□□□□ Math □□□ NULL
- B □□□□ Math □□□——LEFT JOIN □ WHERE □□ INNER □□
- C □□□□□LEFT JOIN □□□□□ WHERE

□□□□□□□□□□□

```sql
-- ❌ WHERE□JOIN □□□□ → NULL □□□□□□ INNER □□
SELECT * FROM students s
LEFT JOIN courses c ON s.student_id = c.student_id
WHERE c.course_name = 'Math';    -- □□□□ Math □□□

-- ✅ ON□JOIN □□□□□ → □□□□□□
SELECT * FROM students s
LEFT JOIN courses c
    ON s.student_id = c.student_id
    AND c.course_name = 'Math';   -- □□ Math □□□ NULL

-- ✅ □□□□□□□□□□→ IS NULL □ WHERE
SELECT * FROM students s
LEFT JOIN courses c ON s.student_id = c.student_id
WHERE c.student_id IS NULL;
```

ON = JOIN □□□□□WHERE = JOIN □□□□□□

PART 04

□□□□□□□□□

```sql
-- □□□□□□□□□□□□
-- students ← enrollments → courses

SELECT
    s.name AS student_name,
    c.course_name,
    e.enrollment_date
FROM students s
INNER JOIN enrollments e ON s.student_id = e.student_id
INNER JOIN courses c ON e.course_id = c.course_id;
```

□□□□□

UNIQUE KEY (student_id, course_id) □□□□□

□□ JOIN

□□ INNER JOIN □□□□□□□□□□□□□

□□□□□ JOIN □□□

□□ 1□□□□ ON

```sql
-- ❌ □□□□
SELECT * FROM students s
INNER JOIN courses c;
-- □□□ = □□□ × □□□
```

□□ 2□□□□□□□□

```sql
-- ✅ □□□□□□□□
CREATE INDEX idx_courses_student_id
    ON courses(student_id);
```

DEBUG FLOW

1. `EXPLAIN` □□□□□ → 2. □□ ON □□ → 3. □□□□□□□□ → 4. JOIN □□ 5 □□□□□□□□

□□□□

DO

- ✓ □□□□□□ `students AS s`
- ✓ □□□□□□□□□ `SELECT *`
- ✓ □□□ LEFT JOIN □□ RIGHT JOIN

DON'T

- × □□ ON □□ → □□□□□□
- × □ LEFT □ INNER → □□□□□□
- × □□□□□□ WHERE → LEFT □ INNER

[illegible]

INNER = [illegible]

[illegible]

LEFT = [illegible]

[illegible] + IS NULL [illegible]

ON vs WHERE

JOIN [illegible] vs JOIN [illegible]

[illegible]

[illegible] ON [illegible]

[illegible]—— [illegible] = LEFT[illegible] = INNER[illegible]

Tables joined.

□□□□□□□□□□□ JOIN □□□

```
$ EXPLAIN SELECT ... FROM a LEFT JOIN b ON ...
inner → left → on/where → index
```